Lab Experiment

Static Routing using CISCO Packet Tracer

AIM:
To configure and verify static routing in cisco packet tracer.

Static Routing:
* static routes are manually added to route routing tables.
* use IP route cmd to add static route
* AD (Administrative distance) decides Preferred route
* lower AD = main preferred route
* higher AD = backup route.

Networks:

Router	Directly connected N/W	other N/W
router0	10.0.0.0, 20.0.0.0, 60.0.0.0	30.0.0.0, 50.0.0.0
router1	20.0.0.0, 30.0.0.0, 50.0.0.0	10.0.0.0, 60.0.0.0
router 2	60.0.0.0, 50.0.0.0	10.0.0.0, 20.0.0.0, 30.0.0.0

Router 0 configuration:

enable
configure-terminal.

IP router 30.0.0.0 255.0.0.0 20.0.0.2 10
IP route 30.0.0.0 255.0.0.0 40.0.0.2 20

IP route 30.0.0.100 255.255.255 40.0.0.2 10
IP route 30.0.0.100 255.255.255 20.0.0.2 20

IP route 50.0.0.0 255.0.0.0 40.0.0.2 10
IP route 50.0.0.0 255.0.0.0 20.0.0.2 20

exit
show ip route static

Main routes: 30.0.0.0 → via 20.0.0.2, AD 10
30.0.0.100 → via 40.0.0.2, AD 10
50.0.0.0 → via 40.0.0.2, AD 10

Router 1 configuration:

enable
configure terminal

IP route 10.0.0.0 255.0.0.0 20.0.0.1 10
IP route 10.0.0.0 255.0.0.0 50.0.0.1 20

IP route 40.0.0.0 255.0.0.0 20.0.0.0 10
IP route 40.0.0.0 255.0.0.0 50.0.0.1 20

exit
show ip route static

Verification:

check routing table: show ip route static.

Test connectivity from a PC: ping <destination

Trace the path: trace <destination-ips>

Test backup route:

i) first, check main route

ii) Delete link blw router0 & router1

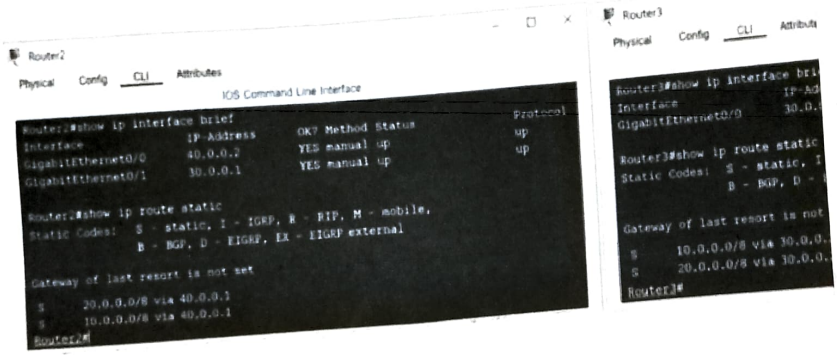
iii) Test again using ping/tracer

Delete a static route:

first check routes: show ip route static

then remove required route:

no ip route <n/w> <subnet-mask> <next-hop>



Result :-

Static routers were successfully configured

Verified in cisco packet Tracer.